CONSERVATION GRANTS FUND

Association of Zoos & Aquariums

2023 AZA CONSERVATION GRANTS FUND

Application Instructions

DEADLINE

This 2023 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT** on **15 March 2023**. An auto-response email with a time-stamp will confirm receipt of the application.

Incomplete applications, or applications received after 5:00 p.m. EDT on 15 March 2023, ***regardless of cause,*** *will NOT be eligible for consideration.* **NO DEADLINE EXCEPTIONS WILL BE MADE.**

MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit *https://www.aza.org/individual-membership.*

FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at *https://www.aza.org/cgf-selection-process*. Applicants will be notified of funding decisions in early September 2023 and funds will be available 2 October 2023. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

APPLICATION GUIDELINES

The application and all supplemental forms are posted online at: *http://www.aza.org/cgf-information-for-applicants/*. These documents contain form fields that must be completed and submitted via email to: *CGFapp@aza.org*. *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. *Only* completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

APPLICATION INSTRUCTIONS AND CHECKLIST: All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

Application: The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled **"Application"** and attached to your Application email.

Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. **NO EXCEPTIONS WILL BE MADE.**

Note that "spell checker" and "track changes" functions cannot be used in this form. AZA strongly recommends that applicants compose the content for their application in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal*. Character limits include spaces.

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list and identify the SAFE species that is (are) the focus of the project. Include a project abstract of 1500 or fewer characters.

- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write "N/A" in the field but *do not leave any form fields blank,* as this may delay processing of your application.

- ☒ **Budget**. Required budgetary information includes:

  - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will *not* be covered by CGF funds.

  - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

Required Attachments: The following must be attached to your Application email:

- ☒ **Curricula Vitae (CV).** A CV is required for each PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, **"CV *Investigator's Last Name*."** *CVs exceeding the maximum length will be cut off at the second page.*

- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI's institution and **each** CI institution *and* collaborating institution must be attached to the Application email as separate documents titled **"SIS *Institution's Name*."** The *Statement of Institutional Support* form can be found with the rest of the Application materials on the AZA website identified above.

  Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions may submit such items as appendices.

Letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form.

- ☒ **Statement of SAFE Species Program Alignment Forms.** Electronically signed *Statements of SAFE Species Program Alignment* from the program leader of relevant SAFE species program(s) must be attached to the Application email as separate documents titled **"SOA *SAFE Species Program Name*."** The *Statement of SAFE Species Program Alignment* form can be found with the rest of the Application materials on the AZA website identified above.

  Letters of support/endorsement from AZA Programs (Taxon Advisory Groups, Species Survival Plans®, SAFE) are **not** accepted, although coordination with AZA SAFE Programs is expected and should be described in Question 1**.** To assure program review of relevant submissions, AZA SAFE or Animal Program Leaders may serve as expert reviewers in the application review process and may express their support for a proposal at that time.

- ☒ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your Application email. Each permit attached to the Application email must be titled **"Permit *#*."** *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2023 to ensure award eligibility*.

- ☐ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for AZA grant funding (i.e., CGF or SAFE granting program) and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the Application email. This attachment must be submitted as a Microsoft Word document titled **"Resubmission Form"** and attached to your Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the Application materials on the AZA website identified above.

Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.

Non-Required Attachments: The following attachments to your Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Application should be included in a separate document titled **"Citations"** and attached to the Application email.

- ☒ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos, publications, or SAFE species program plans. Appendices must be attached to the Application email titled **"Appendix *#*."**

  Note: Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

SUBMISSION INSTRUCTIONS

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA's website at *http://www.aza.org/cgf-information-for-applicants/*.

Proposals must be submitted electronically using the 2023 Application and related forms via a **single email**, **with the proposal title as the subject line**, to *CGFapp@aza.org* no later than 5:00 p.m. EDT on 15 March 2023. Hard copies of any application materials or applications received after this deadline will not be accepted regardless of cause.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2023 Application email **must** include the following attachments

- Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution;
- Statement(s) of SAFE Species Program Support; and
- A Proposal Resubmission form, if applicable.

The Application email may also include citation and appendices attachments.

Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, the person should ensure that an auto-response was received for *each* submission. If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

APPLICATION ASSISTANCE

The CGF website (*http://www.aza.org/cgf-information-for-applicants/*) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 8 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

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2023 AZA CONSERVATION GRANTS FUND
Cover Sheet

Project Number:
For AZA use only

Title of Project: Assessing the Effects of Turtle Racing on Eastern Box Turtle (Terrapene carolina carolina) Health and Movement Behaviors
Amount Requested: $16,826 (round to the nearest whole dollar)
Project Start Date: 10/1/2023 **End Date:** 10/1/2024
(CGF funding is for one year only, beginning no earlier than the award date)

Investigators: Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

1. **Principal Investigator (PI) Name and Title:** [redacted]
   CV Attached? ☒ (required – no more than 2 pages)
   Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (required; available at: *http://www.aza.org/cgf-information-for-applicants/*). ☒
   Individual AZA Member Number: [redacted] (required)
   Institution/Organization: [redacted]
   Statement of Institutional Support Form Attached? ☒ (required)
   Institutional Address: [redacted]
   Phone: [redacted] **E-mail:** [redacted]

2. **Co-Investigator (CI) Name and Title:** [redacted]
   CV Attached? ☒ (required – no more than 2 pages)
   Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: *http://www.aza.org/cgf-information-for-applicants/*). ☒ (required)
   Individual AZA Member Number: [redacted] (required)
   Institution/Organization: [redacted]
   Statement of Institutional Support Form Attached? ☒ (required)
   Institutional Address: [redacted]
   Phone: [redacted] **E-mail:** [redacted]

3. **Co-Investigator (CI) Name and Title:** [redacted]
   CV Attached? ☒ (required – no more than 2 pages)
   Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: *http://www.aza.org/cgf-information-for-applicants/*). ☒ (required)
   Individual AZA Member Number: [redacted] (required)
   Institution/Organization: [redacted]
   Statement of Institutional Support Form Attached? ☒ (required)
   Institutional Address: [redacted]
   Phone: [redacted] **E-mail:** [redacted]

Information about additional co-investigators may be included in a separate document titled **“Additional Co-Investigators”** and attached to your Application email.

Collaborating Institutions or Organizations: A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the "Grant/Matching Support" section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the institutions of the PI and CIs listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the Application email.

Collaborating Institution Names:

St. Louis Zoo *Statement of Institutional Support* Form Attached? ☒ (required)

Murray State University *Statement of Institutional Support* Form Attached? ☒ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled "**Additional Collaborators**" and include it as an attachment to your Application email.

Project Categories (choose two that best describe the project):

☐ Animal / Population Management (*Ex situ*)
☒ Animal Health
☐ Animal Welfare
☐ Communications / Public Relations
☐ Conservation Education / Engagement
☐ Field Conservation (*In situ*)
☒ Research

Identify the SAFE species program(s) this project will benefit: American Turtle

Project Abstract (limit 1,500 characters with spaces):

Eastern box turtles (EBTs; Terrapene carolina carolina) are listed as vulnerable by the IUCN. One threat to EBTs are turtle races, which may influence turtles via removal from the wild, lack of proper care during handling, and, in many cases, release into new territories, which may introduce infectious agents, influence turtle movement and health, and ultimately lead to mortality. Despite these hypothesized threats, there are no published studies to determine the effects of races on box turtles. To obtain scientific data needed to support legislation that could end turtle races, or at the very least inform best management practices at races and ultimately benefit turtle conservation, we aim to evaluate the influence of races on turtle health and movement behaviors. In July 2021, we affixed transmitters to and released 18 turtles we collected from a turtle race onto Clarks River National Wildlife Refuge in western Kentucky. We also captured and placed a transmitter onto eight free-living EBTs from the refuge. We have been tracking the turtles since release in July 2021 and have completed health evaluations three additional times since release. With funding from the AZA, we will continue to monitor the turtles for an additional year. Our data may be used by turtle advocates, such as the Turtle Race Task Force and the AZA SAFE North American Turtle Program, as well as states working towards shifting practices, policies, and educating the public.

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Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance to the SAFE species' program team's priorities and how it supports implementation of the team's 3-year program plan and existing regional and/or global recovery plans. **LIMIT 5,000 CHARACTERS with spaces**

Eastern box turtles (EBTs; Terrapene carolina carolina) are found in central and eastern USA and are listed as vulnerable by the IUCN (van Dijk 2011). One factor that may negatively influence all box turtle species, and yet receives little to no attention, are turtle races. Turtle races are occurring throughout the USA, but are particularly popular in the Midwest (Appendix 1 Fig. 1; Heeb 2020). During these races, multiple species of box turtles, and occasionally other species of North American turtles, are placed into the middle of a series of concentric, chalked rings and the first turtle to exit the last ring is declared the "winner" (Appendix 1 Fig. 2; Heeb 2020). These races are often associated with state, city, or county fairs across the US. It is not uncommon to see hundreds of turtles, generally wild caught box turtles, brought to a race.

Box turtle races are still occurring for several reasons. First, states may lack the ability to enforce regulations if they do have them because of scarce resources (i.e., a lack of funding and workforce) and/or may not have any regulations that would allow policing such races. Second, the public lacks knowledge of how these races may be detrimental to individual turtles and turtle populations and why this matters. Additionally, some people lack empathy for wild animals, which may stem from a lack of education. Regardless of the reasons underlying the continuation of box turtle races in the US, it is clear we lack data to support the need for a shift in policy to better govern races.

Aside from the stress of the race itself, box turtles are often maintained in captivity before the races by people who have limited to no understanding of box turtle husbandry needs. Moreover, many of the box turtles entered in these races end up abandoned at "drop off" locations on site, and then dumped in areas where they are unlikely to survive, or released back at their initial capture location (Appendix 1 Fig. 3). This may have effects on the individual turtle (e.g., stress, changes in movements, and potentially death; Dickens et al. 2010). Moreover, beyond individual turtle health, emerging infectious diseases are a large threat to wildlife generally and chelonians more specifically (Daszak et al. 2000; Adamovicz et al. 2020). Conditions in which turtles are maintained at races are likely to lead to high disease transmission because of close contact between turtles during and after the race. Once a turtle is released, the pathogens it harbors may be introduced into a turtle population, and ultimately may lead to population-level effects. Specifically, box turtles may carry multiple pathogens including ranavirus, adenovirus, herpesvirus, and Mycoplasma spp., which are then transmissible to other turtles (reviewed in Adamovicz et al. 2020) or wildlife (e.g., ranavirus; reviewed in Brunner et al. 2015). Aside from the potential for these turtles to introduce pathogens, these released turtles may be more likely to become diseased in the future, regardless of their initial disease state.

Though turtle races, primarily via disease transmission, may influence individual turtle health and populations of turtles and other wildlife present where they are introduced or reintroduced, to our knowledge there are no studies in the literature on the effects of races on box turtle health or disease epidemiology. Given these gaps, in 2021, we acquired 29 EBTs from a race located in [redacted] On the day of the race, we completed health evaluations of these turtles. After a 2-week quarantine period, we released a subset of turtles (n = 18) at [redacted] Wildlife Refuge ([redacted] following negative ranavirus (frog virus 3) testing results. All turtles weighing > 400 g were affixed with a VHF radio transmitter, and we have been monitoring their health (e.g., body condition, corticosterone hormone levels, various hematology measures [packed cell volume, total solids, and white blood cell counts]), movements, and survival since they were released at the end of July 2021. We have also been monitoring a subset of eight free-living turtles found at [redacted] as a baseline. With AZA funding we aim to continue evaluating the health, survival, and movement of these marked turtles.

Our proposed activities are related to Strategic Objective 10 (Turtle Race Task Force) in the AZA SAFE American Turtle Program Plan and align with program goals 1 and 2. Specifically, our data can be used to help advocates associated with non-profit organizations and states work towards shifting policies and educating the public to reduce the potential negative effects of races on box turtle conservation (American Turtle Program Goal 2). By removing or reducing the occurrence of races, we should see long-term benefits to box turtle populations and other turtle species that have been observed at these races (American Turtle Program Goal 1).

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that the project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

Evidence on the effects of turtle races on the welfare of individual turtles is lacking, and obtaining this information has been identified by the TRTF as a priority. Given the long lifespan of EBTs and the expectation that the effects of anthropogenic impacts may therefore occur over long time periods, we are requesting funding to continue tracking the turtles described above through fall 2024. Our study will complement and support the Turtle Race Task Force (TRFT) in their efforts to evaluate and reduce the potential effects of turtle races on box turtle conservation. For the TRTF to be successful in this, we must obtain data that can be used to support advocacy and outreach efforts. Without data specifically demonstrating the effects turtle races have on box turtles, if present, it will be difficult to garner support for legislation. Using data we have already collected and the data we anticipate collecting in 2023 - 2024 we aim to meet the following objectives:

1) Evaluate effects of turtle races on EBT health, survival, and disease prevalence.
2) Assess how turtle races influence long-term EBT movement behaviors.
3) Disseminate information to organizations and communities to be used for advocating for turtle race regulations and to increase awareness of conservation issues associated with turtle races.

These data can be used to support movements to ban, reduce, or implement best management practices at turtle races, which are currently lacking evidence-based data to provide to organizations and politicians that could help drive these changes. Which approach should be taken will largely depend on the location, as the extent of animal welfare issues vary across race locations, and also the results of ours and similar future studies. Though our aim is to continue monitoring these turtles long-term, by the end of this grant we will have monitored the turtles for slightly over 3 years. Thus, we aim to begin preparation of a peer-reviewed manuscript during the grant period and to submit a paper for publication with these initial results in the first half of 2025. While preparing this manuscript, we will develop a short summary of our results to be shared with organizations, like the TRTF, to help them more easily use our results to help support their efforts to reduce the effects of turtles races on box turtle conservation. Moreover, we will work with education specialists within the [redacted] Zoo [redacted] to develop materials for community outreach and education that incorporate the data we collect.

3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state the research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and why the proposed actions are appropriate for addressing the conservation issue identified. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000 CHARACTERS with spaces**

In July 2021, we collected 29 EBTs from a race held in [redacted] We completed physical exams and collected blood samples and oral and cloacal swabs from each turtle. We quarantined the turtles for two weeks and then released 18 turtles (weight > 400 g) onto a national wildlife refuge in western [redacted] following negative PCR test results for ranavirus (FV3) and the attachment of a transmitter. We also captured eight free-living EBTs from the refuge, collected the same measurements and biological samples, quarantined them for two weeks, and released them with a transmitter at the point of capture. We have been tracking the turtles since release in July 2021 and have completed health evaluations at collection and three times since release. Additional monitoring will occur in 2023 and, with funding from the AZA Conservation Grants Fund, we will continue to monitor the turtles through September 2024. Specifically, we will continue tracking turtles via weekly homing during active months (approximately April through October) and once per month in the winter months. Additionally, we will collect health metrics twice in 2024. These health metrics will be collected in early May 2024 and August 2024.

During a health assessment, we will obtain a blood sample within 3 minutes of handling to obtain baseline levels of corticosterone (Romero et al. 2005; Lamczyk et al. 2022). We will collect the blood sample, which will not exceed 0.5% of body weight, from each turtle using the subcarapacial sinus and a 23- or 25-gauge needle and 3 ml syringe (Jacobson, 1993; Hernandez-Divers, 2006). We will place each blood sample into a lithium heparin tube on ice while in the field. Prior to placing the sample into the lithium heparin tube, we will also make four blood smears per turtle. Following blood sample collection, we will examine each turtle for abnormalities, photograph it, weigh it with a Pesola scale, measure its carapace length and width using calipers, sex each turtle based on characteristics described by Dodd (2001), and evaluate females by digital palpation for the presence of eggs. Additionally, cloacal and mouth swabs will be collected and stored in 1.8 ml cryovials for pathogen testing. If a turtle defecates, we will place a subset of each fecal sample into 10% formalin for long-term storage until we are able to do fecal floats for parasites. The remainder of the fecal sample will be placed into a cryotube for potential calculation of corticosterone. Prior to long-term storage, we will complete hematology diagnostics (e.g., packed cell volume, total solids, and white blood cell counts and differentials). We will also quantify hemoparasite levels from smears made to assess white blood cell counts and differentials. Feces, swabs, and blood samples will be kept cool while in the field and will then be placed in a freezer at -20 °C (feces and blood) or -80 °C (swabs and subset of blood sample) until further analyses are completed.

Analysis of EBT plasma corticosterone concentrations will be done using a commercially available enzyme-linked immunoassay (ELISA) kit (Arbor Assays Detect X Corticosterone). This assay has been validated for use with EBT plasma (A. Darracq, unpublished data). Infectious disease molecular testing will be performed at the ICM molecular lab for ranavirus, adenovirus, herpesvirus, and Mycoplasma spp. following conventional PCR protocols (Wellehan et al. 2004, VanDevanter et al., 1996; Allender et al., 2013; Rebelo et al.2011). Descriptive statistics on disease prevalence will be calculated for the race and CRNWR turtles. Plasma biochemistry values (e.g. uric acid, creatine kinase, cholesterol, and others listed in Appendix 1 Table 1; Deem et al. 2006, 2009) will be obtained from an external reference laboratory.

Data analyses and preservation: All diagnostics will either be completed in the field, the [redacted] Lab (corticosterone), the diagnostics lab at the [redacted] (pathogen screening, PCV/TS, WBC counts, and differentials etc.), or the [redacted] Laboratory ([redacted].com; [redacted]). For long-term storage, plasma and whole blood samples will be maintained frozen at -80 °C. We will use area corrected, optimally weighted autocorrelated kernel density estimation to calculate 95 % home and 50 % core ranges for each turtle (Fleming et al. 2015, 2018). Additionally, we will calculate the total distance moved (TDMs) and the mean daily distance moved (MDM) for each turtle following Otten et al. (2021). To assess how these metrics change with increasing time since release, we will calculate the 95% home and 50% core range, TDM, and MDM for each turtle annually and overall. We will evaluate differences in the annual and overall 95% home and 50% core ranges, TDMs, and MDMs and the various health metrics described above between race and CRNWR turtles using general or generalized linear mixed models. Generalized models will be used when data are non-normal and mixed models will be used to account for non-independence of data points when analyses include the same metric collected from an individual turtle multiple times. For the health metrics, we will also consider the interaction between sampling period and source (i.e., race versus CRNWR turtle) to account for the potential effects of time since release on these measures.

4. If AZA has provided previous funding for this or a related project through its grants programs, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

   N/A

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

- October 2023: Funding awarded.
- October 2023 – September 2024: Field data collection, entry, and sample processing.
- August – October 2024: Finish sample processing and communication of results via preparation of a peer-reviewed manuscript (likely submitted in the first quarter of 2025) and sharing data with stakeholders and collaborators.

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

[redacted] is a Certified Wildlife Biologist® and Associate Professor [redacted]. [redacted] will lead the field data collection and conduct the enzyme-linked immunosorbent assays to quantify corticosterone. [redacted] also mentors the graduate student on this project. [redacted] has a background in statistics and writing and will work with her graduate student to analyze the data and complete first drafts of manuscripts associated with this project.

[redacted]eem is a wildlife veterinarian/epidemiologist and the director of the [redacted] Zoo [redacted] will serve as the veterinary advisor for turtle health evaluations and will work with [redacted] to ensure personnel involved in data collection (e.g., the graduate student and technician) are properly trained in the acquisition of samples.

[redacted] is the field and lab technician for the [redacted] [redacted] will be involved in training the personnel involved in data collection specifically related to hematologic and serologic evaluation of samples and the biomaterial storage for molecular diagnostics.

Both [redacted] and [redacted] have experience conducting health evaluations in the field and are experts on sample collection, processing, diagnostics, and storage. Given this, their expertise on analyses and interpretation of the health related data will be vital to this project. [redacted] and [redacted] will help with manuscript editing and statistical analyses and interpretation as needed.

[redacted] is a Research Associate with the [redacted] with experience in molecular diagnostics, phylogenetics, and virology. [redacted] has been conducting regular diagnostic testing of chelonians as part of the [redacted] programs and will conduct pathogen testing of the turtles in this study. Though [redacted] will also help with overall manuscript editing and statistical analyses and interpretation as needed, her expertise in the analysis of disease prevalence data will be essential to our proposed project.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed). **LIMIT 3,000 CHARACTERS with spaces**

Our data from health screenings and the long-term monitoring of turtles post-release will be the first to elucidate the potential effects of races on turtle welfare. Additionally, though we are focused on individual turtles for this project, we can develop hypotheses for how these effects may influence turtle populations from where the turtles were removed, as well the populations into which turtles are introduced or reintroduced. For instance, currently we hypothesize these turtle races may lead to disease transmission both at the race and within populations into which the turtles are released. However, we have no data specific to races to support these claims. We will also provide data to organizations to help implement evidence-based policies and regulations as we work to stop, reduce, or develop best management practices for these races and to develop educational materials for the public on why turtle racing is a threat for turtle conservation.

In the short-term, we will have achieved our first two objectives with this project if we are successful in monitoring turtle movements and survival and collecting the health assessment metrics and disease data from the turtles that we proposed above. Additionally, we aim to begin preparations of at least one peer-reviewed manuscript during the CGF granting period. This manuscript will likely be submitted in early 2025, following input on the manuscript from all collaborators. We will also work with the Turtle Race Task Force and education specialists within conservation-related organizations nearby races to begin developing outreach and education materials. Though full dissemination of our results (Objective 3) will take longer than the one-year grant period, our work on this component of the project will be ongoing beyond the completion of the grant. We will consider ourselves successful during the granting period if we are able to begin a first draft of our peer-reviewed manuscript and to provide summary results to stakeholders to be used in their advocacy and education efforts.

In the long-term, we will work with the TRTF and other relevant stakeholders to determine further needs to support efforts to reduce the effects of races on box turtle conservation. As stated above, in conjunction with other studies being completed and efforts by the TRTF, we expect to see a decrease in turtle race activities in the USA. Regulatory changes take time, but we do anticipate an increase in state-level regulations involving turtle races over the next 10 – 20 years. Thus, this project and other research we aim to complete in the future will support the broader goal to increase implementation of best management practices at races and/or turtle race regulations at the state level by advancing our understanding of how turtle races influence box turtles.

8. Explain and identify the means and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

We will share these data with relevant stakeholders and, in conjunction with these stakeholders, develop priority needs for further studies needed to provide the data-based evidence required to support the creation of best management practices and/or regulations for turtle races. Our data will form the basis for developing further questions that may need to be addressed as we currently lack any data on the influence of turtle races on turtle health, movements, or survival.

Regulatory changes can take time and depend on a state's current political environment. Thus, though the long-term goal is to reduce the occurrence of turtle races, our data will also be used to inform best management practices for turtle races to minimize the negative effects these races are having on turtles. We will work with the Turtle Race Task Force as they develop these to incorporate our data on health, movements, and survival of turtles at races as they compare to turtles sampled in their natural environments.

In conclusion, the results of this study will give the TRTF and other related organizations the science-based evidence they need to advance their goals to minimize the negative effects of races on box turtle conservation. We will prepare an article for the AZA Connect and, in conjunction with work we began in summer 2021, we anticipate multiple peer-reviewed manuscripts will result from these data. We will continue to present our results at regional and national conferences.

9. Have all the necessary permits been obtained for the project?

   a) ☒ Yes ☐ No ☐ Not Applicable

   If yes, list the permits here and attach one copy of each to your Application email. Attachments should be titled **"Permit #."**

   [redacted] permit and [redacted] collection permit #S[redacted]

   b) If the necessary permits have not yet been obtained, please explain why. Pending permits must be obtained and submitted to AZA by 15 June 2023.

This project has already been approved by the [redacted] University IACUC and a continuation protocol is in progress. We do not anticipate any issues with receiving this. [redacted] currently has a permit from [redacted] Wildlife Refuge and the [redacted] Department of Fish and Wildlife Resources. These permits will be reapplied for at the end of 2023, but the current permits are attached.

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

N/A

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

If funded, we will have tracked the turtles we are currently monitoring for three full summers post-release and, even if we are unable to continue monitoring, will consider our project successful. However, our aim is to continue monitoring these turtles for as long as possible. Thus, we will pursue additional funding from other internal (e.g., The [redacted] Institute, small internal faculty and student grants at [redacted], and the [redacted] Conservation grant) and external (e.g., The [redacted] and other non-profit organizations interested in box turtle conservation) sources to continue our monitoring efforts beyond 2024.

12. **Past AZA Grant Support.** If Principal or Co-Investigators listed on the Cover Sheet have received previous AZA grant support (i.e., CGF or SAFE granting program), complete the following (required).

   a) Name of Investigator: [redacted] Year of Award: 2018 Amount of Award: 18,536

   Project Title: [redacted]

   Were all required reports (annual/final) submitted?

   ☒ Yes ☐ No ☐ Not yet due / Other (briefly explain)

b) Name of Investigator: [redacted] Year of Award: 2013 Amount of Award: 20,834

Project Title: [redacted]

Were all required reports (annual/final) submitted?

☒ Yes ☐ No ☐ Not yet due / Other (briefly explain)

Note: If additional awards need to be included, add their information to a separate document titled "**Past AZA-Funded Projects**" and include it as an attachment with your Application email.

CONSERVATION GRANTS FUND

Association of Zoos & Aquariums

2023 AZA CONSERVATION GRANTS FUND

Project Budget

BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x $25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over $500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

CGF *does not* fund the following: Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation goals.

CGF *does not* fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.

Note: Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's Scientific Advisory Committee may recommend that the award amount be reduced or refunded.

Note: If additional budget items need to be included, add this information in a separate document titled **"Additional Budget Items"** and include it as an attachment with your Application email.

Budget Items Form:

Budget Item	Method of Calculation (e.g., 2 persons x 10 days x $25/day)	**CGF-** Requested Amount	**Non-CGF** Requested Amount (if applicable)	Date Expected from **Non-CGF Source** (if applicable and not yet received)	Name of **Non-CGF Source** (e.g., Grant Fund, Matching)	Total Expense (CGF+Non-CGF)
Technician	15 hrs/week x $10/hour x 37 weeks	$5,250	$			$5250
Fringe	17 weeks @ 0.55% and 16 weeks @ 8.2%	$	$235.425		[redacted]	$235.425
Transportation	37 weeks x 200 mi/week x $0.55/mi	$4,070	$			$4070
PCR pathogen screening	2 swabs/turtle x 24 turtles x 2 health screenings x $32/sample	$3,072	$			$3072
Plasma biochemistry	24 turtles x 2 health screenings x $24/sample	$1,152	$			$1152
PCV, TS, Differentials	24 turtles x 2 health screenings x $18/sample	$864	$			$864
CORT assays	24 turtles x 2 health screenings x $16/sample	$768	$			$768
Consumables		$750	$			$750
Receiver	Wildlife materials TRX 1000s	$900	$			$900
		$	$			$
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	TOTAL PROJECT BUDGET (round to nearest whole dollar)	**$16,826**	**$235.425**			**$17,061.425**

BUDGET JUSTIFICATION

State why each item listed in the Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over $500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

Our budget includes a request to fund one field assistant for 15 hours per week at $10 per hour over 37 weeks. These hours are based on our current tracking efforts, as we are able to track all turtles within 15 hours each week. Fringe benefits (e.g., Workman's compensation), which will be covered by the [redacted]e, are calculated per [redacted] guidelines at 8.2% in the summer months and 0.55% in the fall and spring semesters. Mileage is requested for travel to and from the field site estimated at 7400 miles traveled and $0.55/mile. The budget also includes funding for diagnostic testing and supplies needed to perform and store samples associated with health examinations. PCR pathogen screening is estimated at $32 per sample and 144 samples (2 swabs per turtle x 24 surviving turtles x 2 health evaluations), plasma biochemistry at $24/sample and 48 samples, packed cell volume, total solids, and differentials at $18/sample and 48 samples, and corticosterone $16/sample for 48 samples. $750 is requested to purchase consumables including swabs, lithium heparin vials, microcentrifuge tubes, and other materials needed for field data collection and processing the samples in the lab. Funding is being requested to purchase a receiver to replace our current one, which will be in the incorrect frequency range once we change transmitters. Additionally, our current unit is not functioning well and the receivers we do have access to that are in the correct frequency are associated with classes, making it difficult to access them when needed. The receiver would continue to be used to track these turtles in the future by [redacted] at [redacted] University.